

Getting Started with Google Colab: A Beginner-Friendly Guide



BOARD MEMBERS:

V. Jarana (Chairperson), Prof. Y. Ballim, M. Fakir, M. Govender, M. Matolong,
Dr V. Mthethwa, M. Mulcahy, J. Newton, Dr C. Render, Prof. A. Van Zyl, Dr T. Dlamini (CEO)

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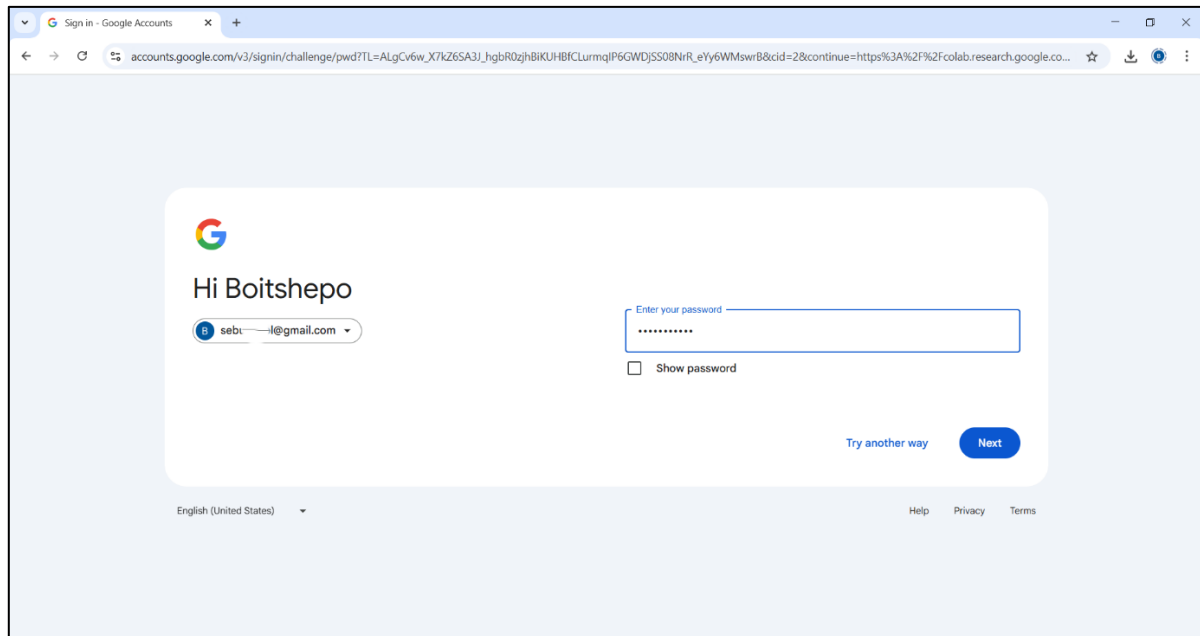
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Introduction

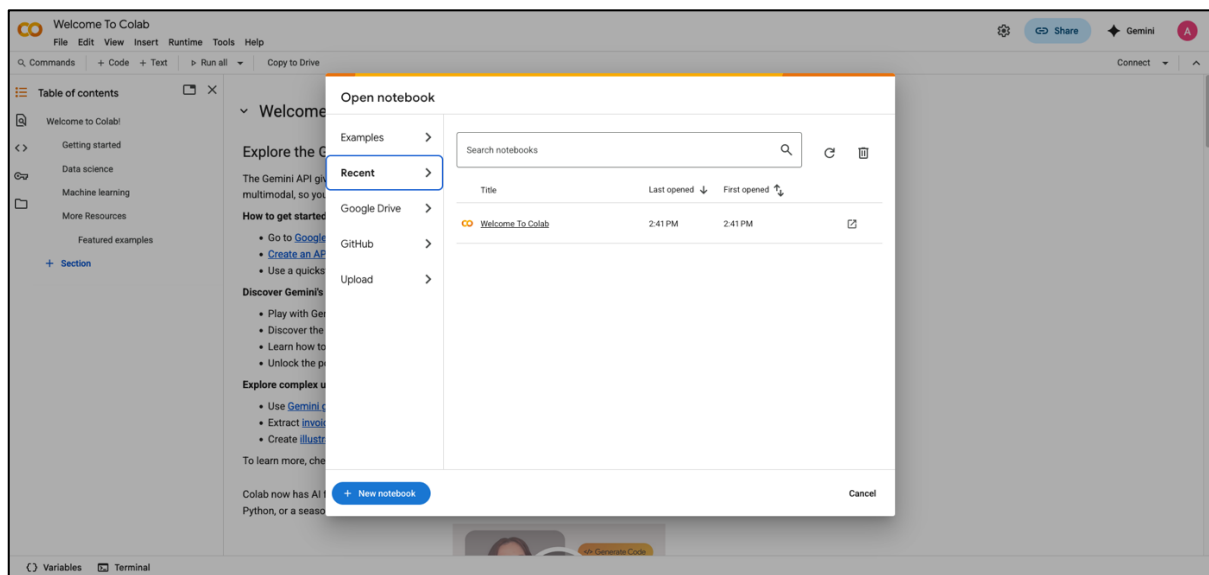
Google Colab is a free, cloud-based coding environment that supports python and popular data science libraries. This guide helps you open, run, and save notebooks easily. Google Colab requires a Google account. It is ideal for learning, sharing, and collaborating

Getting Started with Google Colab

- Visit: <https://colab.research.google.com>
- Log in using your Google account.
 - Note: If you do not have a Google Account, you will have to create one.

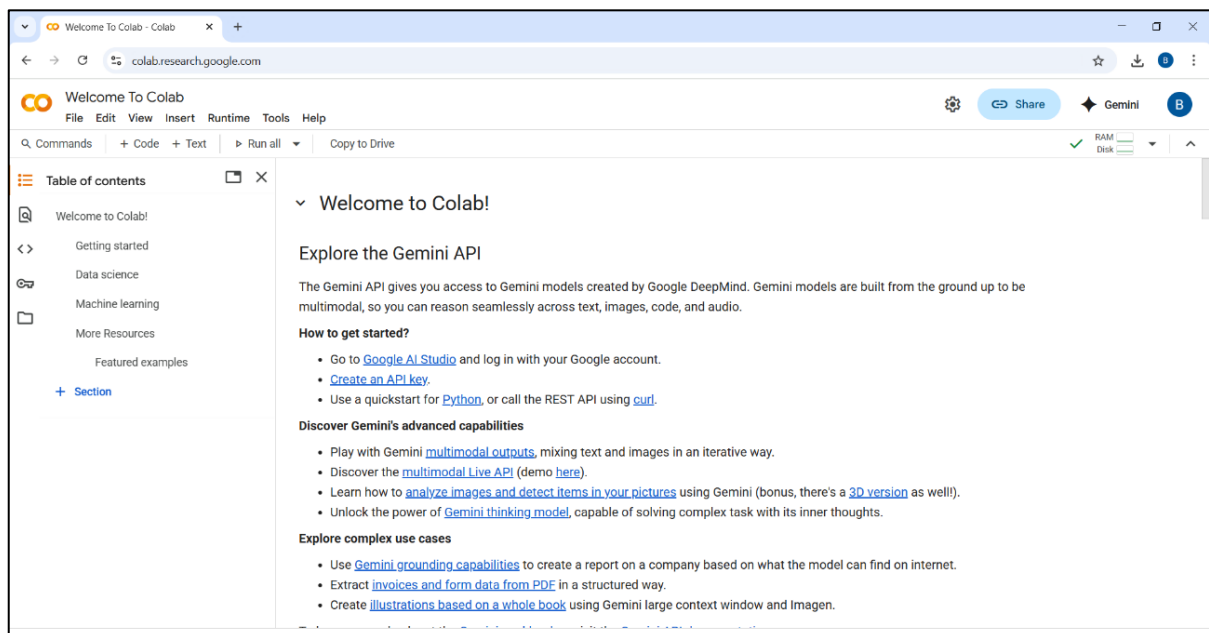


- After logging in, you will see the home page for Google Colab



Google Colab Home Page

The Google Colab Home Page contains various useful links to help you get started. You can explore these resources.

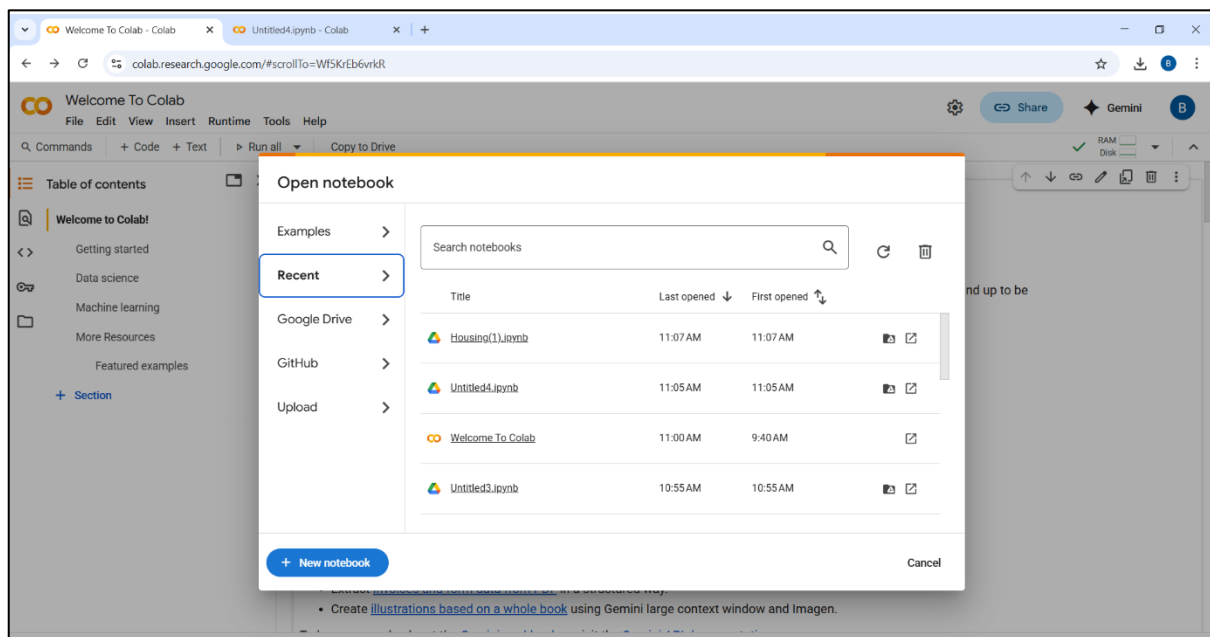


Starting a Notebook

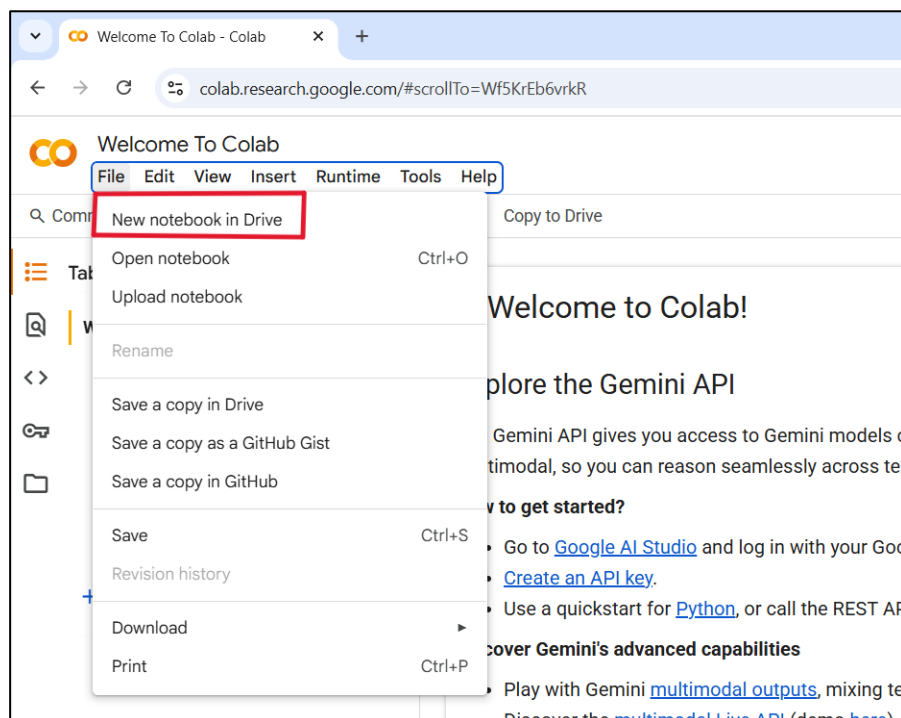
To create or access a notebook, you need to choose one of the below:

- Create a 'New notebook'
- Upload a notebook from your computer
- Open an existing notebook from Google Drive or GitHub

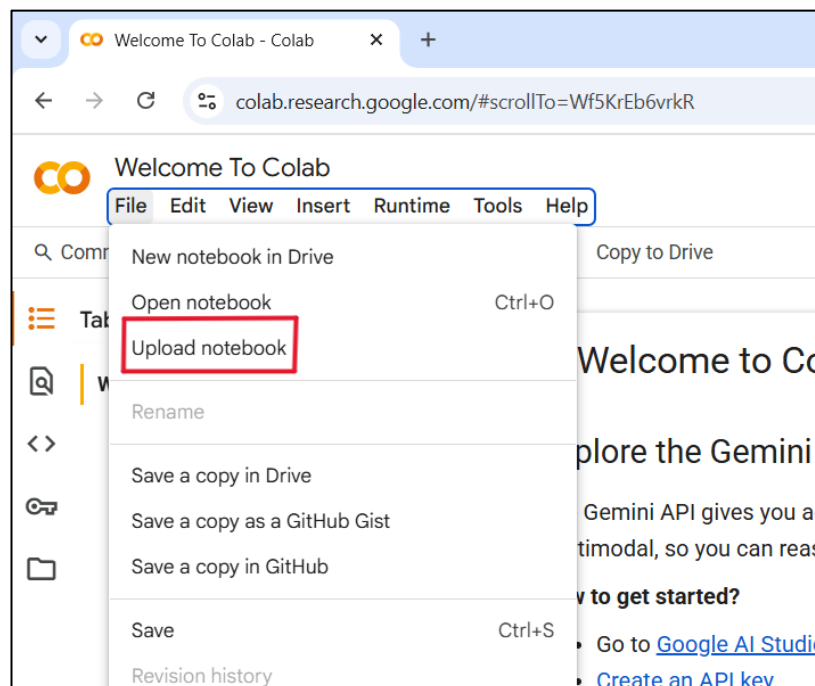
- Open recently accessed notebooks



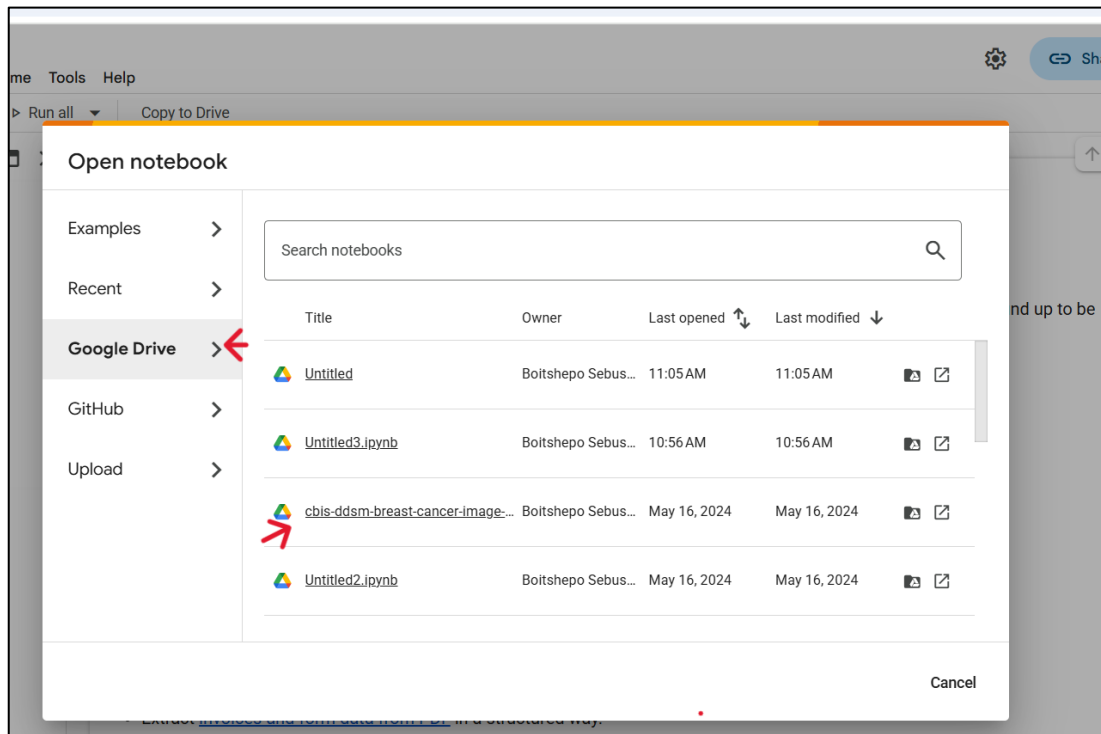
- You can also create a new Notebook by clicking on 'File' and selecting 'New Notebook in Drive'.
- This will create a new newbook in your Google Drive:



- To upload an existing notebook, you can click on 'File' and select 'Upload Notebook':



Various options to upload a notebook are presented to upload a notebook from Google Drive, GitHub or from your computer:



- Once uploaded, the notebook is opened and shown:

```
import os
import numpy as np
import pandas as pd
import cv2
import seaborn as sns
import matplotlib.pyplot as plt
import matplotlib.image as plt
import tensorflow as tf
from sklearn.model_selection import train_test_split
from sklearn.utils import class_weight
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Conv2D, MaxPooling2D, Flatten, Dense, Dropout
from tensorflow.keras.utils import to_categorical
from tensorflow.keras.callbacks import ModelCheckpoint, EarlyStopping
from PIL import Image
```

/opt/conda/lib/python3.10/site-packages/scipy/_init_.py:146: UserWarning: A NumPy version >=1.16.5 and <1.23.0 is required for this version of SciPy (detected version 1.24.3)

warnings.warn(f"A NumPy version >=({np_minversion}) and <({np_maxversion})")

/kaggle/input/cbis-ddsm-breast-cancer-image-dataset/jpeg/1.3.6.1.4.1.9590.100.1.2.426013102412536473840491086280519425818/1-223.jpg

/kaggle/input/cbis-ddsm-breast-cancer-image-dataset/jpeg/1.3.6.1.4.1.9590.100.1.2.16568542011655175430071162144126831780/2-089.jpg

/kaggle/input/cbis-ddsm-breast-cancer-image-dataset/jpeg/1.3.6.1.4.1.9590.100.1.2.16568542011655175430071162144126831780/1-111.jpg

/kaggle/input/cbis-ddsm-breast-cancer-image-dataset/jpeg/1.3.6.1.4.1.9590.100.1.2.116437756113164247142539498550746659420/1-051.jpg

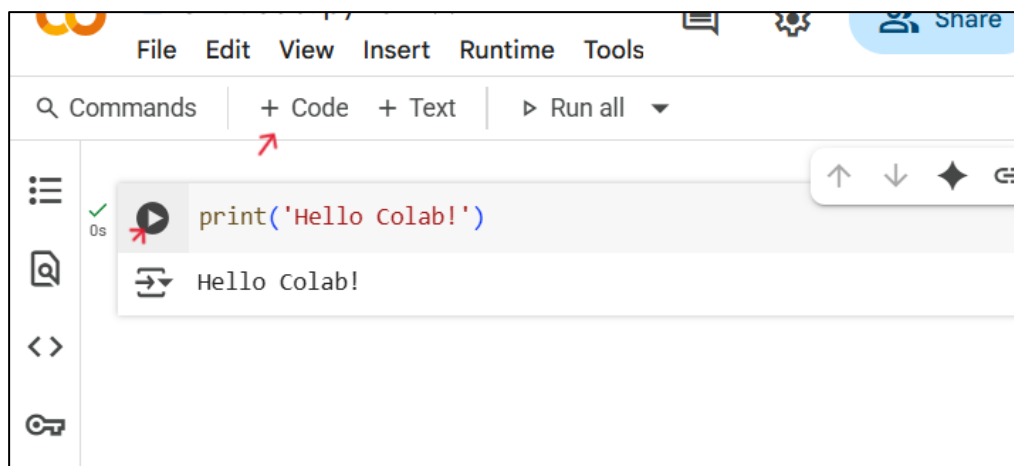
/kaggle/input/cbis-ddsm-breast-cancer-image-dataset/jpeg/1.3.6.1.4.1.9590.100.1.2.116437756113164247142539498550746659420/2-203.jpg

/kaggle/input/cbis-ddsm-breast-cancer-image-dataset/jpeg/1.3.6.1.4.1.9590.100.1.2.265953197912973660327069359641506533925/1-288.jpg

/kaggle/input/cbis-ddsm-breast-cancer-image-dataset/jpeg/1.3.6.1.4.1.9590.100.1.2.205670217013556255028452747763741627478/1-213.jpg

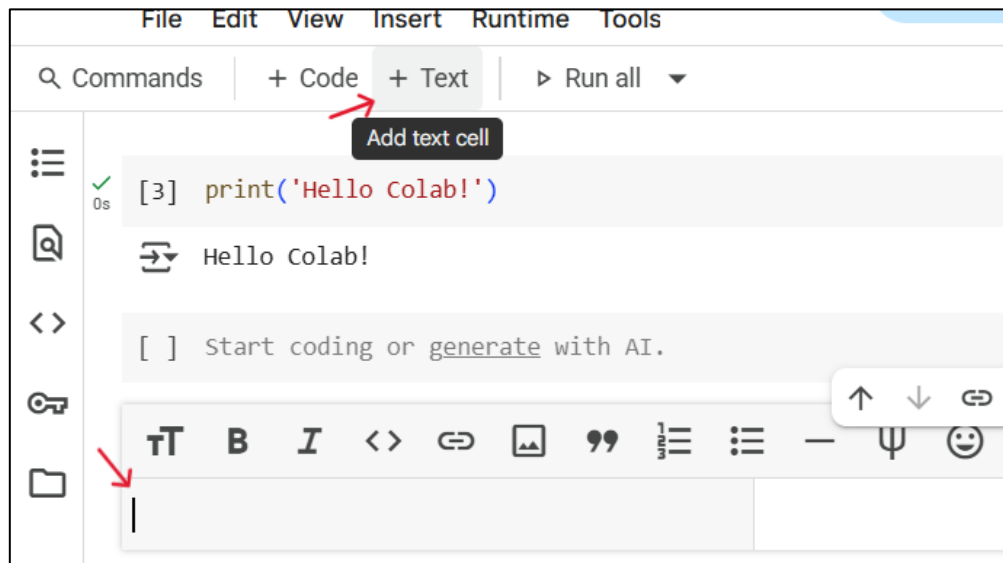
Writing and Running Code

- Click on '+Code' to add a new code cell that will allow you to start writing Python Code.
- To run a code cell, you need to use the  button or Shift + Enter



Using Text Cells (Markdown)

- Use text cells for formatting, adding explanations, or instructions:



- You can mix text and code cells for clarity.

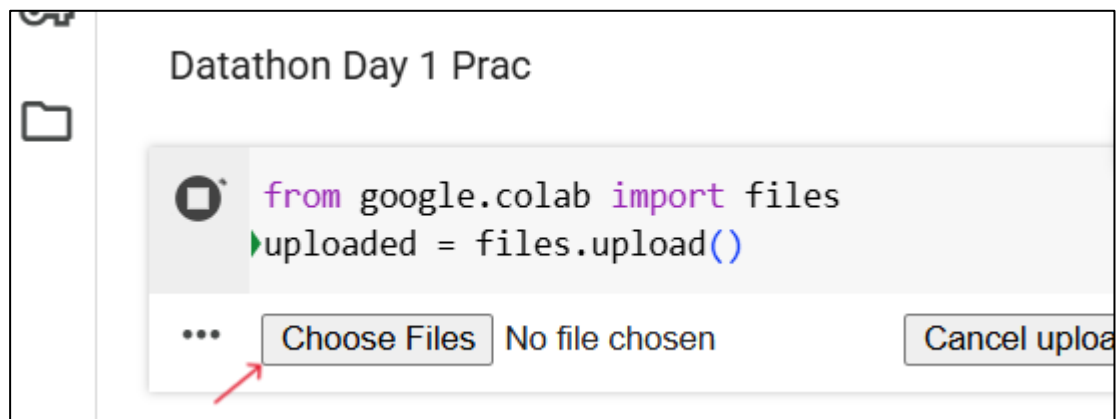
Uploading/Downloading Files

Upload using Code

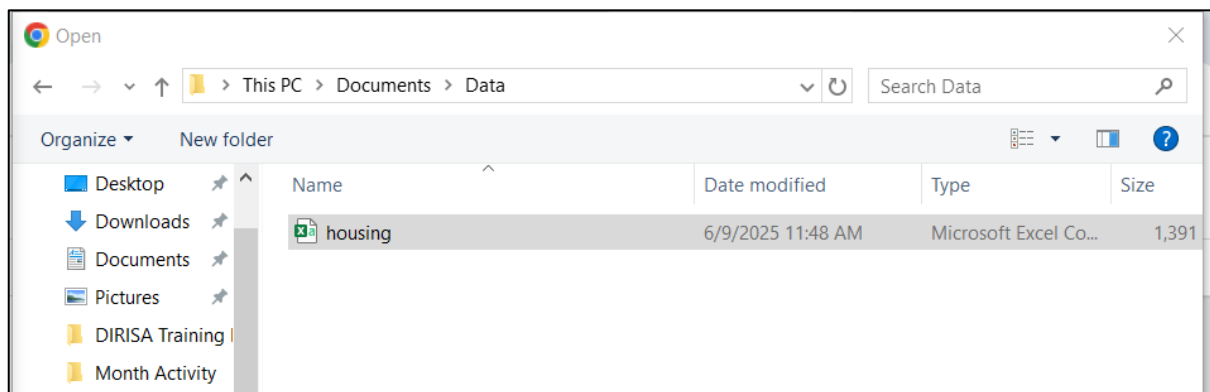
- To **upload files** using code, you can type the below code in a code cell:

```
from google.colab import files
```

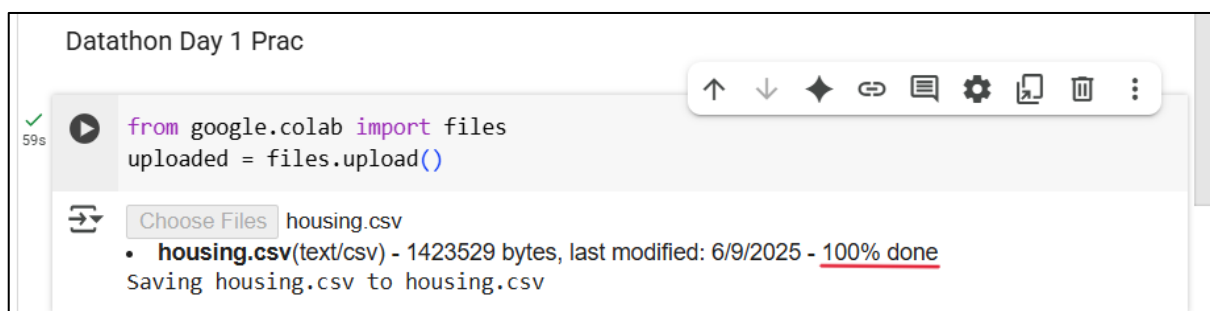
```
uploaded = files.upload()
```



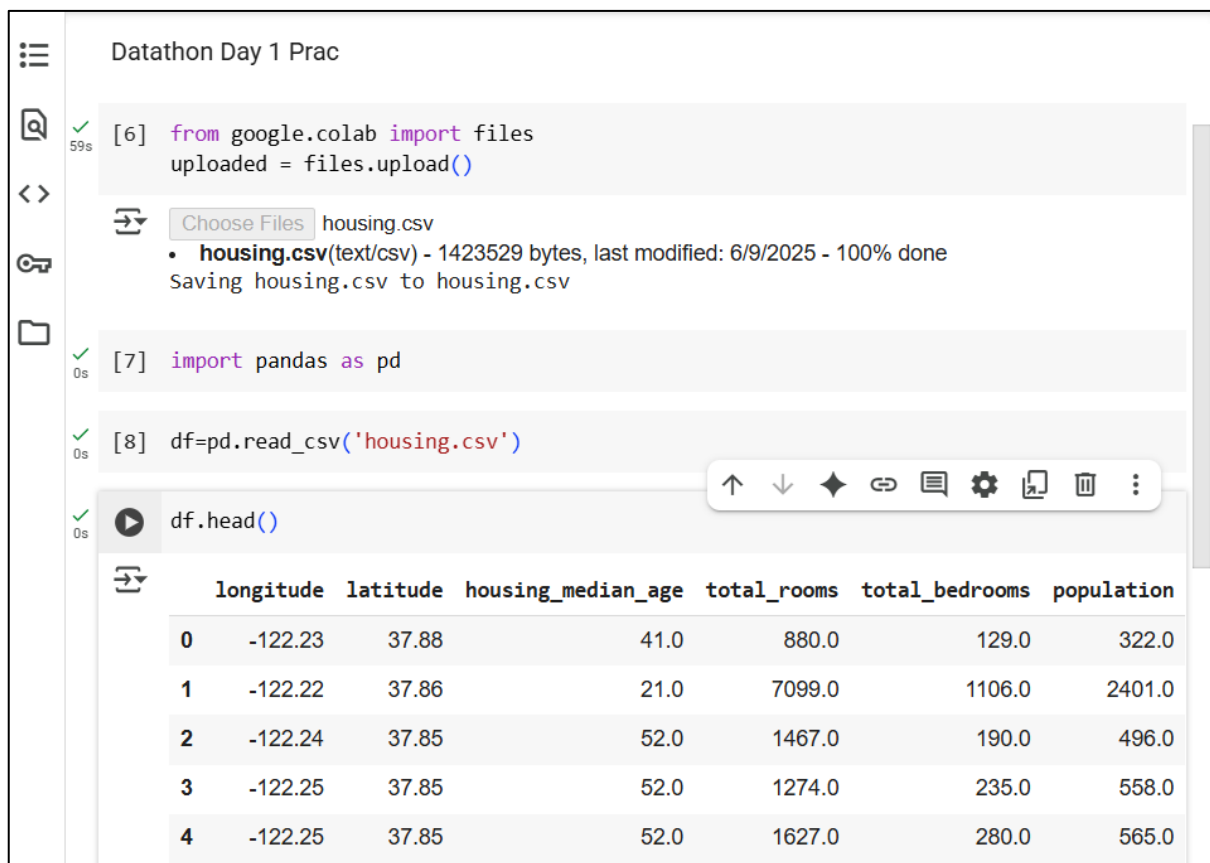
- Select the files by clicking on 'Choose Files':



- Wait for the files to upload:



- You can now use the uploaded file:



Mount Google Drive to access saved files

- To mount your Google Drive to access saved files, you need to write the following code in your code cell:

```
from google.colab import drive  
drive.mount('/content/drive')
```

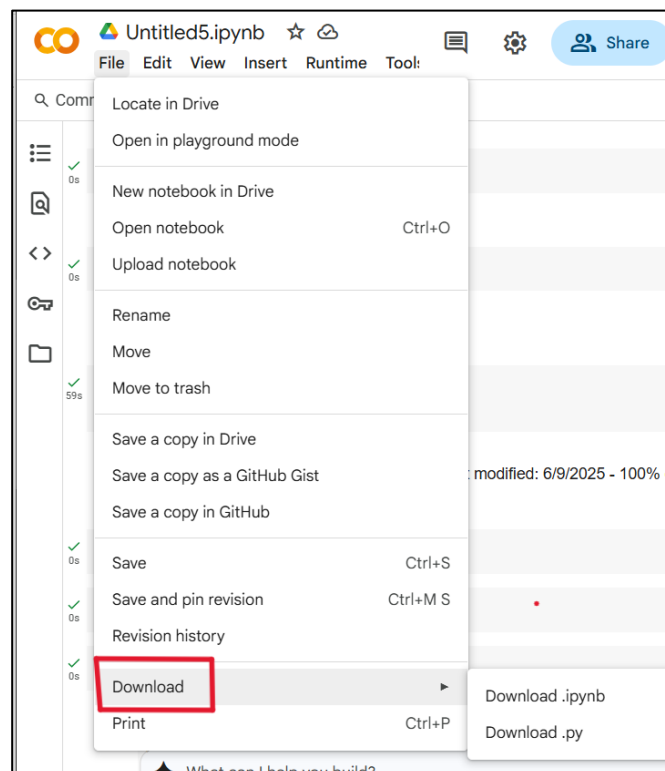
- You will be required to accept the permissions that appear.
- If the mount is successful, you will see the below message:



Download Notebook

To download a notebook:

- Click on 'File'
- Select 'Download'
- Choose download type as .ipynb or .pdf



Saving Work

- Colab notebooks **auto-save** changes to your Google Drive in real-time
- To save manually:
 - Click on 'File'
 - Select 'Save a copy in Drive' or 'Save a copy as .ipynb'
- **Best Practices:**
 - Name your notebook clearly: File → Rename
 - Save different versions using File → Save a copy to avoid overwriting
 - Keep notebooks in organized folders in your Google Drive

Troubleshooting & Help

- **No output?** Make sure the runtime is connected (check top right)
- **Error messages?** Read carefully and debug one line at a time
- **Kernel crash?** Runtime → Factory Reset
- **Resources:**
 - Colab Help Docs
 - Python & Data Science Tutorials on YouTube

You're now ready to explore Google Colab and start coding!

Basics

Python Basics:

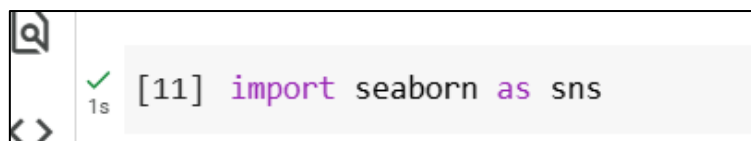
- Variables: `x = 5`
- Lists: `colors = ['red', 'blue']`
- Loops: `for c in colors:`
- Functions: `def hello(): print("Hi")`

Install and Import Libraries

Common Libraries:

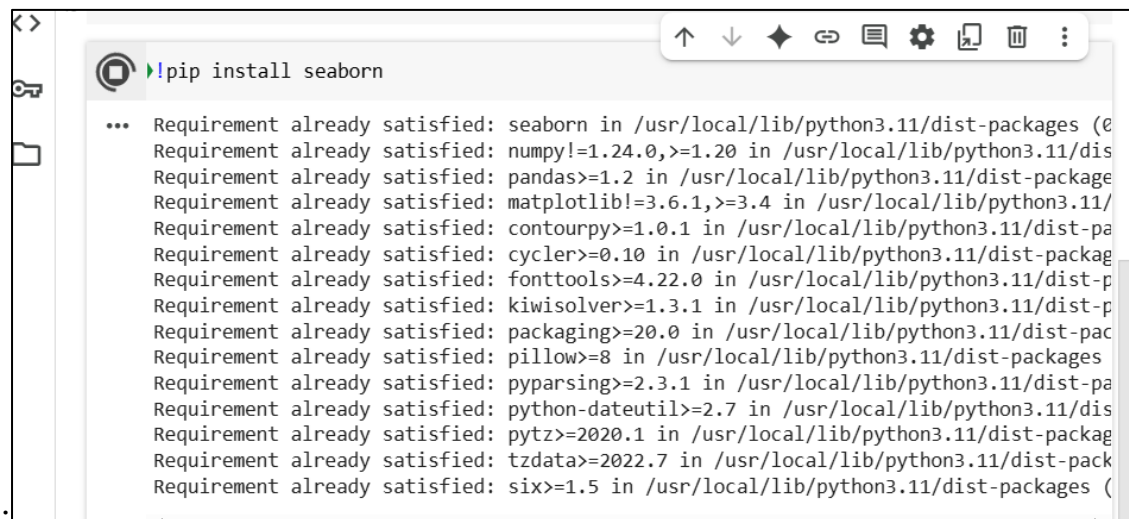
- pandas: For data manipulation
- numpy: For numerical operations
- matplotlib, seaborn: For data visualization

This is how to **import** a library:



```
[11] import seaborn as sns
```

Install using `!pip install` command if not already available:



```
!pip install seaborn

... Requirement already satisfied: seaborn in /usr/local/lib/python3.11/dist-packages (0.12.2)
Requirement already satisfied: numpy!=1.24.0,>=1.20 in /usr/local/lib/python3.11/dist-packages (1.26.4)
Requirement already satisfied: pandas>=1.2 in /usr/local/lib/python3.11/dist-packages (2.2.2)
Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in /usr/local/lib/python3.11/dist-packages (3.9.2)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.11/dist-packages (1.3.0)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.11/dist-packages (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.11/dist-packages (4.53.0)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.11/dist-packages (1.4.7)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.11/dist-packages (24.1)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.11/dist-packages (10.4.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.11/dist-packages (3.2.0)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.11/dist-packages (2.9.0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.11/dist-packages (2024.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.11/dist-packages (2024.2)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.11/dist-packages (1.17.0)
```

Practice Activity

Try this:

- Open a new notebook
- Import pandas and load a CSV file
- Use Markdown to describe the data
- Plot a simple chart using matplotlib
 - Tip: You would have to import matplotlib